

Phase 3: Project Design Phase

System & UI Design Document

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System Architecture Flow

The system follows a standard client-server Machine Learning architecture, as illustrated below:

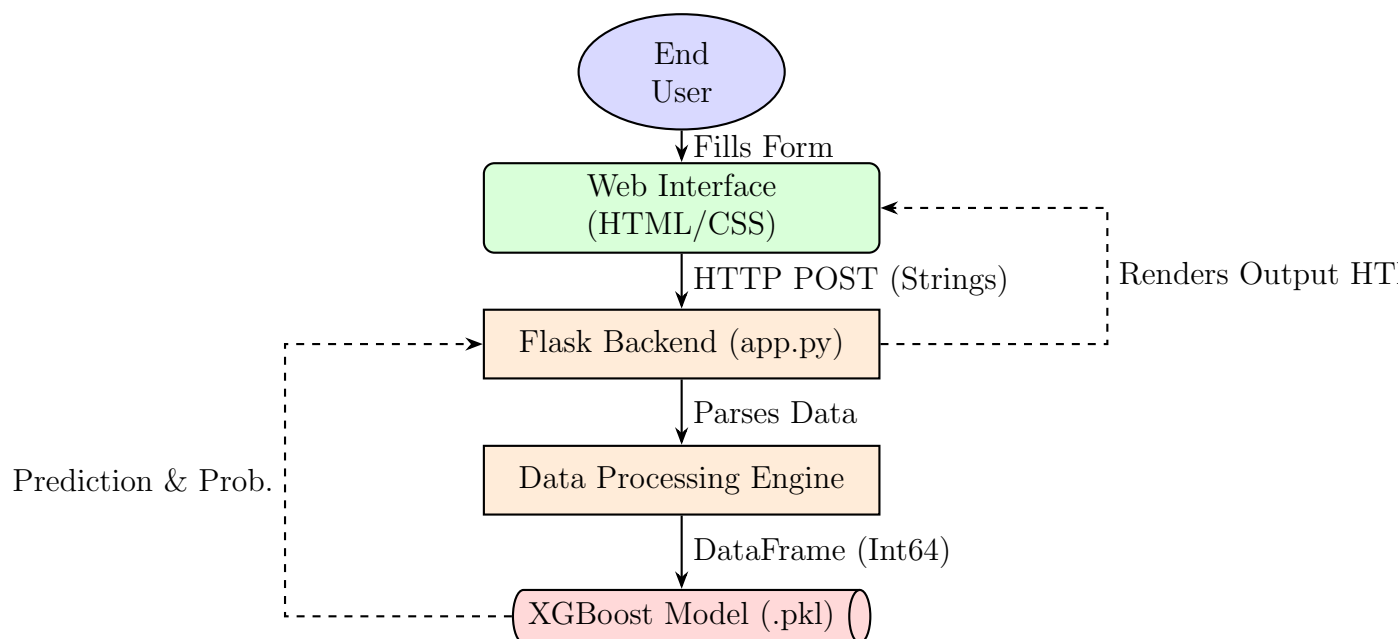


Figure 1: SmartLender System Architecture Flow

Data Flow Logic

When user data arrives via HTTP POST, it is in string format. The Flask app executes a strict sequence of transformations to ensure XGBoost receives the exact `int64` format it was trained on, using `model_columns.pkl` to prevent feature mismatch errors.

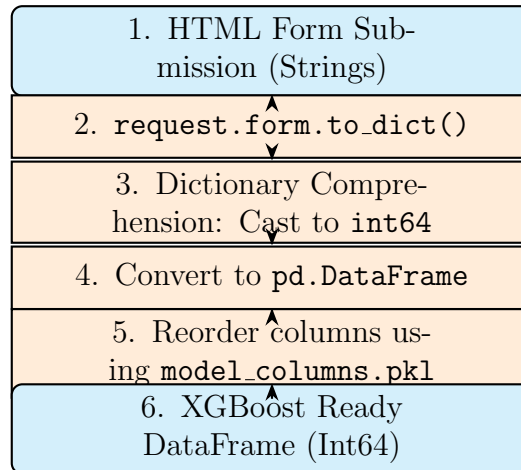


Figure 2: Data Preprocessing and Transformation Flow

Entity-Relationship (ER) Diagram

The dataset follows a single-entity structure, as all features belong to one unified loan application record. The diagram below illustrates the `LOAN_APPLICATION` entity and its grouped attributes.

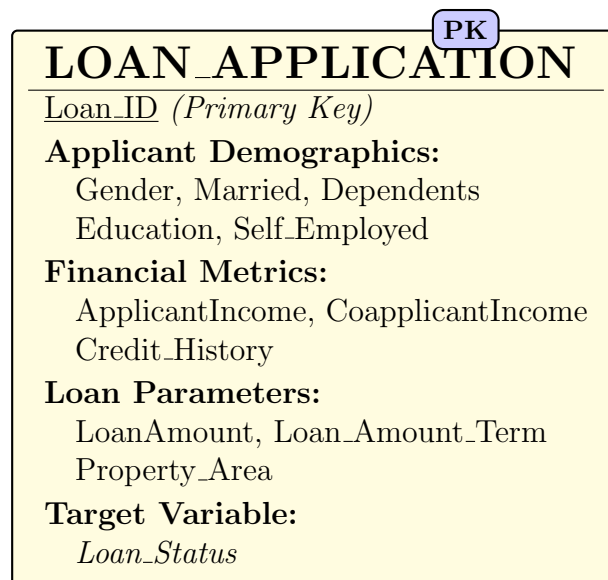


Figure 3: Entity-Relationship Diagram for Loan Dataset

User Interface Mockups

- **Home Page (home.html):** A clean landing page displaying project details, the tech stack used, and a prominent green “Start Application” button that routes to the form.
- **Input Page (input.html):** A structured HTML form containing dropdowns for categorical variables (mapped to integer encodings, e.g., Male=1, Female=0) and number inputs for continuous variables like **ApplicantIncome** and **LoanAmount**.
- **Output Page (output.html):** A focused result page displaying the loan status (APPROVED in green text / REJECTED in red text) and the model’s confidence percentage.